

Machine learning helps to strongly reduce future warming uncertainty

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The observed warming of global mean surface temperature has been used to reduce uncertainty in future climate change and impact projections, but the information embedded in the spatial pattern of warming remains largely untapped. Here, we use machine learning to uncover spatially resolved emergent constraint relationships between 1971–2020 warming trends at individual grid cells and future global mean warming in a large collection of climate model simulations. This approach identifies key tropical and polar regions, with clear physical interpretations, where historical warming most effectively constrains future outcomes. Including the observed pattern information reduces the error variance in future global warming projections by about 70%, while commonly used constraints based solely on the global mean trend reduce that uncertainty by about 48%. These refined projections imply a higher likelihood of early exceedance of Paris Agreement thresholds. For example, the likelihood of exceeding 2 °C by mid-century under the SSP3-7.0 scenario is approximately 80%, compared to 70% when constrained by observed global mean trend alone. Our study highlights the potential of machine learning to uncover physically interpretable emergent constraints that improve future climate projections.

Anthropogenic global warming is profoundly affecting environmental and societal systems^{1,2}, causing extensive and multifaceted damage^{3–8}, with disproportionate impacts on vulnerable underdeveloped regions^{9–11}. Addressing these challenges demands effective mitigation and adaptation strategies^{2,12,13} that ultimately depend on reliable projections of future climate change and impacts. There has been substantial effort over recent decades to reduce uncertainties in climate change projections by improving climate models^{14–17} and developing emergent constraints from instrumental observations^{11,13,18–35} and paleoclimate data^{36–39}. Progress from these efforts has enabled the Intergovernmental Panel on Climate Change (IPCC) to publish assessed global warming projections in its Sixth Assessment Report (AR6)

with lower uncertainty than previously possible^{1,40}, offering a critical foundation for understanding and preparing for future climate risks. A central pillar of these improved projections is the integration of estimates constrained by observations of global mean warming^{1,18–20}.

Nevertheless, the pronounced spatial heterogeneity of temperature changes, arising from anthropogenic forcings, regional climate feedbacks, and variations in internal variability, raises concern that the observed global mean warming, as an integrated metric, may not provide the most effective possible constraint. Spatial patterns of projected warming in sea surface temperatures^{41,42} and near-surface air temperatures⁴³ among climate models have been identified as important contributors to uncertainty in both regional and global warming

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projections. While some aspects of sea surface temperature variations have been used to inform future warming projections^{22,44,45}, the potential of the full global historical warming fields to refine future warming projections, and to reveal where historical observations most effectively constrain these projections, has remained largely unexplored.

By applying a machine learning method to a large collection of simulations from phase six of the Coupled Model Intercomparison Project⁴⁶ (CMIP6; Supplementary Table S1), here we uncover spatially resolved emergent constraint relationships between historical warming at individual grid cells and future global mean warming. This analysis identifies critical tropical and polar regions where historical warming provides additional constraint information beyond the global mean warming alone. Applying the spatial constraint relationships to the observed warming field reduces the error variance of future global warming projections by about 70%, nearly 1.5 times the reduction achieved by a constraint based on the observed global mean warming alone. This improvement translates into more certain local climate projections, facilitating more robust and targeted climate adaptation planning at impact-relevant scales.

Results

Future warming constrained by past warming field

We train a machine learning model to simultaneously constrain future global mean near-surface air temperature (GSAT) warming for near-term (2021–2040), mid-term (2041–2060), and long-term (2081–2100) periods relative to preindustrial 1850–1900 under a given Shared Socioeconomic Pathway (SSP) scenario⁴⁷, using a map of 1971–2020 grid-cell warming trends as input (“Methods”). This historical period is chosen because the effective radiative forcing and warming during this time results primarily from anthropogenic greenhouse gas emissions¹³, making the associated warming particularly relevant for informing warming under future emissions scenarios, which is also dominantly driven by increases in greenhouse gases⁴⁸. Training a single machine learning model to constrain multiple projection periods maintains model parsimony relative to using separate models for each period, while exploiting the physical coherence of global warming across time horizons under a given emissions pathway.

The warming field proves to be substantially more effective in reducing uncertainty in future global warming projections than the global mean (Fig. 1a). Under the SSP5-8.5 fossil fuel-intensive emissions scenario⁴⁷, which projects extreme risks that natural and societal systems may face this century, the 5–95% uncertainty ranges constrained by the observed warming field derived from HadCRUT5 (ref. 49) are 1.38–1.98 °C for the near-term, 2.14–2.97 °C for the mid-term, and 3.83–5.63 °C for the long-term. These constrained warming ranges are roughly 47% narrower than their unconstrained counterparts, representing an error variance reduction of about 70%, and about 25% narrower than projections constrained by the global mean warming trend using a linear regression-based hierarchical emergent constraint method^{13,50} (HEC; see Supplementary Information). They are also much narrower than the IPCC assessed ranges¹ (Supplementary Fig. S1a). Averaged across all projection periods and scenarios considered, the machine learning warming field constraint reduces projection error variance by more than 70% relative to unconstrained projections, which is nearly 1.5 times the reductions achieved by the HEC global mean constraint (~48%; Supplementary Fig. S1b). These results, constrained by the HadCRUT5 warming field, are nearly identical with those obtained when other observational datasets^{51–53} provide the constraint (Supplementary Fig. S2).

Overall, the constrained projections suggest less future warming than the unconstrained projections, consistent with previous findings that many CMIP6 climate models are overly sensitive to greenhouse gas emissions relative to observational evidence^{54–58} and with estimates derived from the HEC global mean constraint (red vs. orange

bars in Fig. 1a). Nevertheless, the warming reduction is smaller than suggested by the IPCC assessed projections (Supplementary Fig. S1a). The SSP1-2.6 scenario presents an exception, for which the constrained projections show comparable warming to unconstrained estimates. This likely reflects a known limitation that historical warming during periods of rapid transient response to forcing is less effective at constraining future warming under stabilization or overshoot scenarios^{13,29}.

The machine learning warming field constraint narrows uncertainty by reducing the upper uncertainty bounds and raising the lower uncertainty bounds of the projections (purple vs. orange bars in Fig. 1a), unlike the global mean constraint that primarily reduces upper bounds (purple vs. red bars in Fig. 1a). The elevated lower bounds imply that warming thresholds set by the United Nations Paris Agreement⁵⁹ could be reached earlier than suggested by the HEC global mean constraint (dashed horizontal lines in Fig. 1a) or IPCC assessed projections (Supplementary Fig. S1a). Under the SSP2-4.5, SSP3-7.0, and SSP5-8.5 emissions scenarios, the machine learning warming field constraint yields exceedance probabilities of 0.62, 0.8, and 0.97 for 2 °C warming by mid-century (2041–2060), larger than 0.58, 0.7, and 0.9 from the HEC global mean constraint. Despite projecting less warming than the unconstrained projections, the warming field constraint still indicates substantially earlier 2 °C exceedance due to its elevated lower uncertainty bounds. These results carry important implications for global climate policy, underscoring the urgency for more immediate mitigation.

Climate model-based evaluation

We use a cross-validation test to verify that the machine learning warming field constraint reliably produces projections with reduced bias and error variance (measured by root-mean-square error; RMSE) without being overconfident. For each emissions scenario, simulations from each climate model are used in turn as a validation sample that includes both historical warming trend fields and corresponding future warming outcomes. The machine learning model is trained on the remaining climate simulations, excluding those from the model providing the validation sample, and applied to historical fields in the validation sample to generate constrained future warming estimates, which are then compared with the corresponding warming simulations in the validation sample to evaluate constraint performance (see details in Methods).

We find a RMSE of 0.33 °C between the constrained warming estimates and the verifying model simulations, calculated by pooling all projection periods and emissions scenarios (Fig. 1b; see Supplementary Table S2 for results for individual periods and scenarios). Approximately 87% of the validation simulations fall within the constrained 5–95% uncertainty ranges (statistics in Fig. 1b), and about 66% fall within the constrained 16–84% ranges, both close to the expected values of 90% and 68%, respectively (see Methods for coverage estimation). These coverage rates, which remain consistent across projection periods and scenarios (Supplementary Table S2), indicate no evidence of overfitting and that the machine learning warming field constraint is not overly conservative. Compared to the HEC global mean warming constraint, it yields lower RMSE and narrower uncertainty ranges (Fig. 1b, c and Supplementary Table S2, S3), benefiting from the additional constraint information contained in the spatial patterns of warming.

Physical interpretations

To understand how the warming pattern information enhances the constraint provided by the global mean trend, we analyze local warming sensitivities in climate models, defined as the ratio of local to global mean warming (Fig. 2a, b for future and historical periods, respectively). These sensitivities exhibit spatial patterns that are nearly invariant over time and across emissions scenarios, consistent with

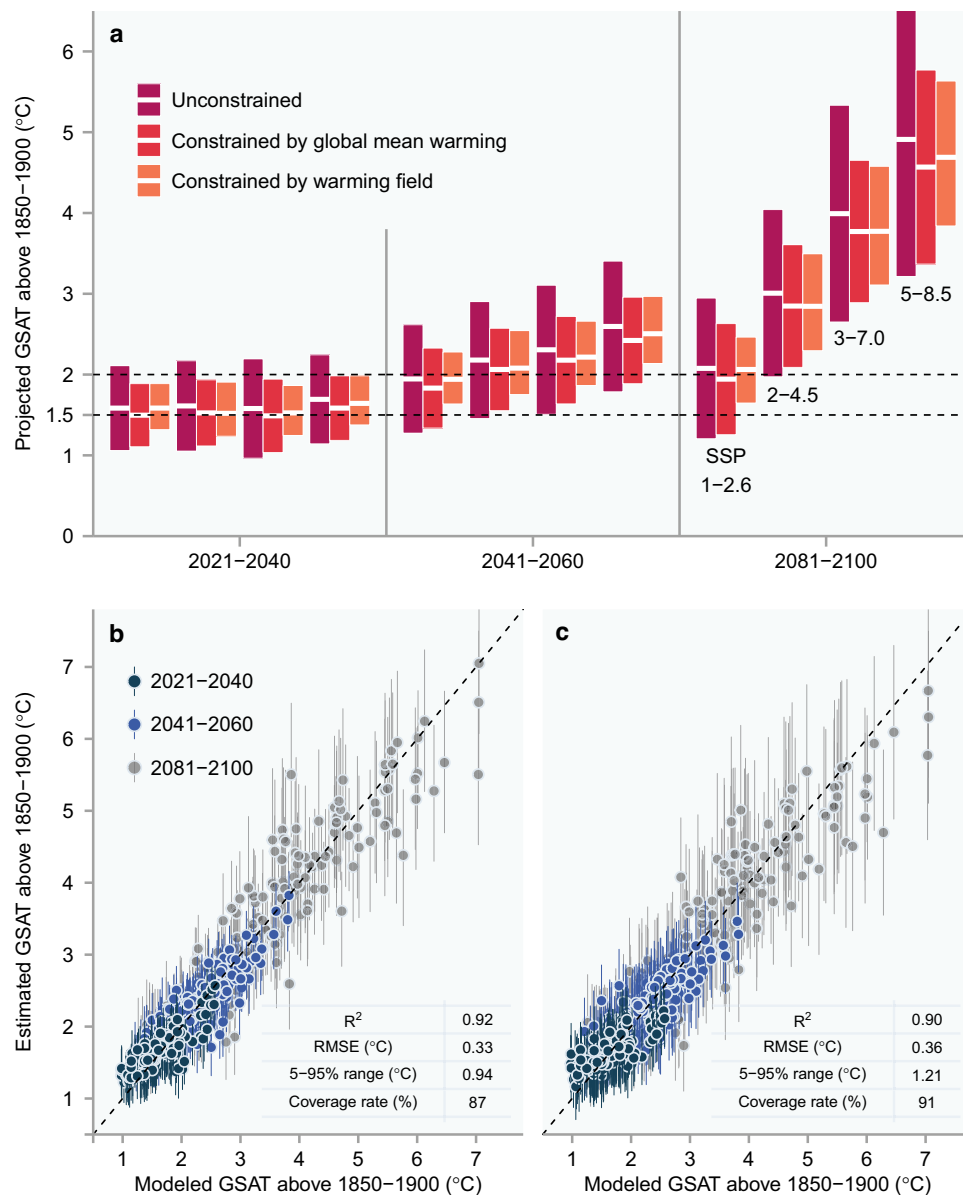


Fig. 1 | Constrained global warming projections. **a** Global mean near-surface air temperature (GSAT) changes in 2021–2040, 2041–2060, and 2081–2100 relative to 1850–1900 under the Shared Socioeconomic Pathway (SSP) 1–2.6, SSP2–4.5, SSP3–7.0, and SSP5–8.5 emissions scenarios, based on unconstrained projections (purple), projections constrained using hierarchical emergent constraint (HEC) based on global mean warming (red), and projections constrained by machine learning warming field (orange). The white horizontal lines indicate the means of the projections, while the bars represent their 5–95% uncertainty ranges. Dashed black lines denote the 1.5 °C and 2.0 °C global warming thresholds. **b**, **c** Leave-one-model-out cross-validation results of the machine learning warming field constraint (**b**) and HEC global mean warming constraint (**c**). Scatter points represent GSAT changes in

2021–2040 (black-blue), 2041–2060 (blue), and 2081–2100 (gray) relative to 1850–1900 for different emissions scenarios. The x-axis shows the climate model projections, and the y-axis shows the corresponding constrained projections. Vertical bars denote the 5–95% uncertainty ranges of the predictions. Bottom-right values present the coefficient of determination (R^2) and the root-mean-square error (RMSE) between scatter points, the range of 5–95% uncertainty ranges, and the coverage rate of the 5–95% uncertainty ranges encompassing the corresponding climate model projections. All performance statistics are calculated by pooling all projection periods under different emissions scenarios. The dashed black line represents the 1:1 reference line.

previous findings⁶⁰. This temporal invariance is a necessary prerequisite that allows historical sensitivities to meaningfully inform future sensitivities, which, in turn, affect global warming projections (Supplementary Fig. S3), forming a constraint chain from historical local warming sensitivity to future global warming. Consistent with this expectation, we find statistically significant positive inter-model correlations between historical local warming sensitivities and future global warming projections across extensive regions, including the Arctic, Tibetan Plateau, South and Southeast Asia, the Amazon, and the Indian and Southern Oceans (stippling and contours in Fig. 2a, b). These regions are known to be affected by pronounced feedback

processes, such as ice-albedo feedback in the Arctic⁶¹, cloud-radiative feedbacks associated with tropical ocean warming^{62,63}, and upwelling-related ocean heat uptake in the Southern Ocean⁶⁴. They also largely coincide with areas where historical warming trends exhibit stronger correlations with future global warming projections (Fig. 2c), suggesting that warming sensitivities in these regions, reflecting the integrated influence of regional feedbacks, provide additional constraint information beyond the global mean trend alone.

The machine learning model allows us to assess the relative efficiency of historical warming trends at individual grid cells to future global warming projections (“Methods”). The 100 grid cells with the

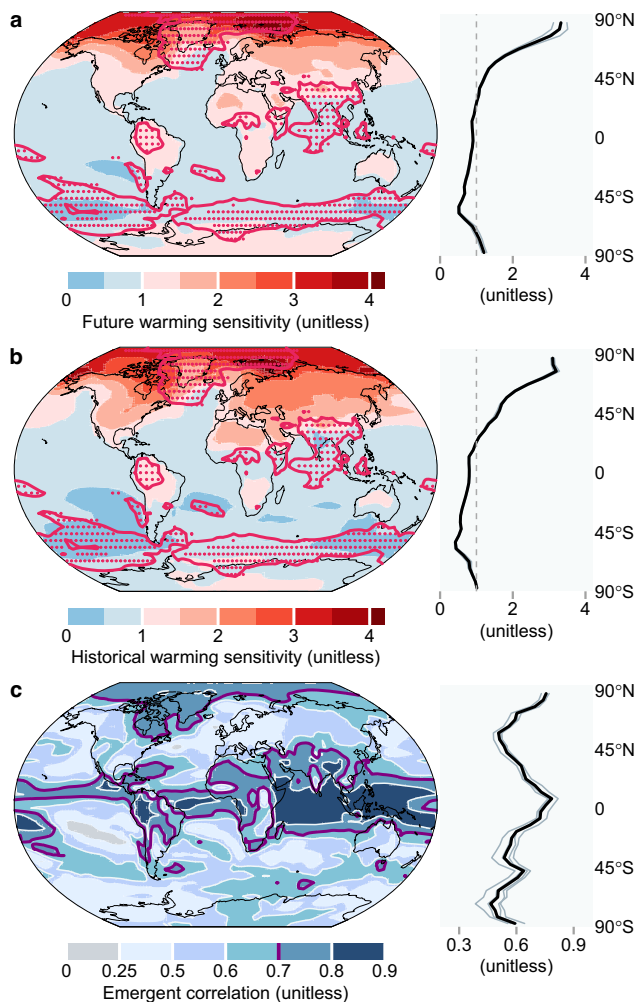


Fig. 2 | Constraint information from historical warming patterns. **a** Mean local warming sensitivities in the projections of annual mean near-surface air temperature changes relative to 1850–1900, averaged across different future periods and emissions scenarios. **b** Counterparts of **(a)**, derived from the 1971–2020 simulations from the ensembles corresponding to the future projections. The historical simulation ensembles vary across emissions scenarios (Supplementary Table S1). Stippling in **(a)** and **(b)** marks locations with statistically significant correlations (5% level) between 1971–2020 local warming sensitivities and future global warming projections in at least 10 of the 12 scenario periods (i.e., 3 projection periods $\times$ 4 emissions scenarios), while contours mark the 0.2 mean of these correlations. **c** Mean correlation coefficients between 1971–2020 grid-cell warming trend simulations and future global warming projections, averaged across different periods and scenarios. For all panels, black curves on the right indicate the zonal means of the mapped values, while gray curves represent averages across projection periods for each emissions scenario, with one gray curve per scenario.

highest average contributions across emissions scenarios are shown in Fig. 3a. While their specific locations and relative contributions vary across scenarios, their overall spatial distribution remains stable (Supplementary Fig. S4) and robust to the choice of machine learning algorithm (Supplementary Fig. S5). The most influential grid cells are concentrated within or near the regions where historical warming sensitivities and trends exhibit strong correlations with future warming projections (pink and purple contour lines in Fig. 3a, respectively). Importantly, the machine learning model does not treat all grid cells within these regions as equally informative, but instead preferentially weights a small subset. In general, grid cells with relatively low uncertainty in warming sensitivities, indicating higher climate model consistency in representing underlying feedback processes,

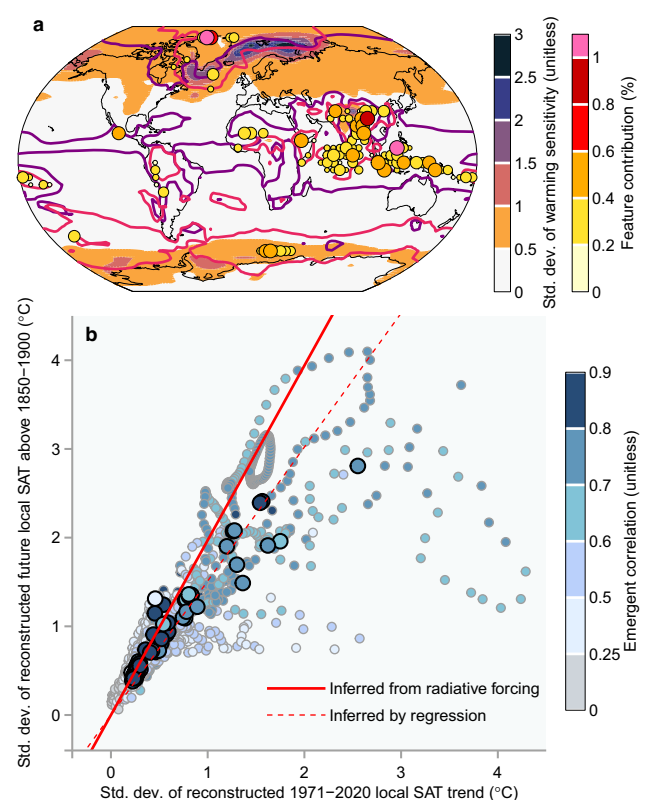


Fig. 3 | The machine learning warming field constraint interpretation. **a** Points mark the 100 most influential grid cells identified based on the mean normalized Prediction Value Change (PVC) scores from machine learning models trained for generating constrained projections for different future periods under different emissions scenarios (“Methods”). Point sizes are proportional to their respective PVC scores. Filled contours show the average, across emissions scenarios, of the standard deviations in historical local warming sensitivities, calculated separately for each scenario using historical 1971–2020 simulations with corresponding projections (Supplementary Table S1); note that the historical simulation ensembles differ across emissions scenarios. The red and purple contours replicate those shown in Fig. 2. **b** Scatter plot of the mean standard deviations of 1971–2020 warming trends (x-axis) and global mean near-surface air temperature projections (y-axis) for different periods and scenarios, based on warming amounts that have filtered to isolated the contributions from the respective dominant first modes of warming uncertainty (Methods). Point colors correspond to the correlation coefficients shown in Fig. 2c. The 100 most influential grid cells marked in panel **(a)** are highlighted with larger points. The solid and dashed red lines (both passing through the origin) represent the ratio of mean radiative forcing driving warming for different periods under different scenarios relative to the forcing during 1971–2020 and the linear regression fitted to the top 100 influential grid cells, respectively. Corresponding results for individual emissions scenarios are shown in Supplementary Fig. S6.

contribute more effectively (filled contours in Fig. 3a). Conversely, grid cells marked by large uncertainty are down-weighted, such as those associated with poorly constrained sea ice-related feedbacks in parts of the Eurasian Arctic, limiting the effectiveness of historical warming sensitivities and trends as constraints.

The standard error of simulated warming responses is expected to scale linearly with the underlying effective radiative forcing when temperature sensitivities to external forcing remain approximately constant with warming levels (“Methods”). This simple scaling relation offers an independent, physically informed interpretation of how the machine learning model identifies the most influential grid cells. The scatterplot of standard deviations in 1971–2020 grid-cell warming simulations versus future projections, based on warming amounts that

have been filtered to isolate the contributions from the respective dominant first modes of warming uncertainty (“Methods”; Fig. 3b). We find that historical versus future warming variations at the most influential grid cells align closely with a line whose slope approximates the ratio of effective radiative forcings driving the future and historical warming (solid red line in Fig. 3b; see corresponding results for individual scenarios in Supplementary Fig. S6), consistent with expectations from the linear forcing-response scaling relation. This indicates that the machine learning model prioritizes grid cells where temperatures exhibit stable sensitivities to radiative forcing across warming levels, thereby enabling historical warming sensitivities and trends to effectively constrain future global warming projections, in agreement with the interpretation outlined above.

In summary, multiple lines of evidence support the physical validity of the machine learning constraint relationships, which in addition to empirical cross-validation, further reinforces confidence in our constrained global warming projections. These include (i) the near temporal invariance of local warming sensitivities under different emissions scenarios, which enables historical information to effectively inform future responses; (ii) the strong influence of warming sensitivities in the identified critical regions on future warming projections, allowing constraints on these sensitivities to efficiently reduce uncertainty in future global warming; (iii) the preferential weighting by the machine learning model of grid cells that satisfy both conditions above and exhibit high inter-model agreement in warming sensitivities, and (iv) the consistency of the most influential grid cells with a linear forcing-response scaling under approximately stable warming sensitivity.

Supporting projections of warming impacts

The constrained global warming projections can be used to refine climate impact projections that scale with global warming¹³ (“Methods”). As an illustration, Fig. 4 presents the constrained projections of mid-term changes in grid-cell annual mean near-surface air temperatures relative to pre-industrial 1850–1900 under the SSP3-7.0 scenario, which is viewed by many as being more compatible with current climate mitigation policies than other scenarios^{65,66}. Southern Hemisphere and most Northern Hemisphere low- to mid-latitude land areas are projected to warm by ~1–2 °C above recent (1995–2014) levels, while most Northern Hemisphere high-latitude land areas are expected to warm by ~2–5 °C (Fig. 4a), both slightly below unconstrained estimates. The projection error variance is below 0.25 °C² across nearly all land areas (Fig. 4b), corresponding to a 5–95% uncertainty range of ±0.8 °C under a Gaussian assumption. Compared to unconstrained projections, the local error variance over land is reduced by an average of 43%, with 90% of land areas having reductions of 15% to 60% (Fig. 4c). Similar uncertainty reductions occur under other scenarios (Supplementary Fig. S7), although they are less for SSP1-2.6, where weaker forcing results in lower spread in unconstrained projected changes relative to internal variability and thus less opportunity to reduce projection spread through the application of constraints. These more precise local projections can enhance climate impact assessments at impact-relevant levels, supporting more targeted adaptation planning.

Discussion

The spatial patterns of sea surface temperature (SST) changes can influence regional and global warming through, for example, cloud-radiative feedbacks^{41,42,62,63}. In particular, the cooling pattern in the eastern tropical Pacific SSTs during recent decades, presumed to be internally generated, has dampened the observed global warming trend, possibly biasing constrained projections downwards^{22,42}. Nevertheless, this SST cooling pattern is weak during 1971–2020 compared to the 1981–2020 period often considered, and its teleconnection influence on near-surface air temperature is largely confined to the eastern tropical Pacific cold tongue (Supplementary

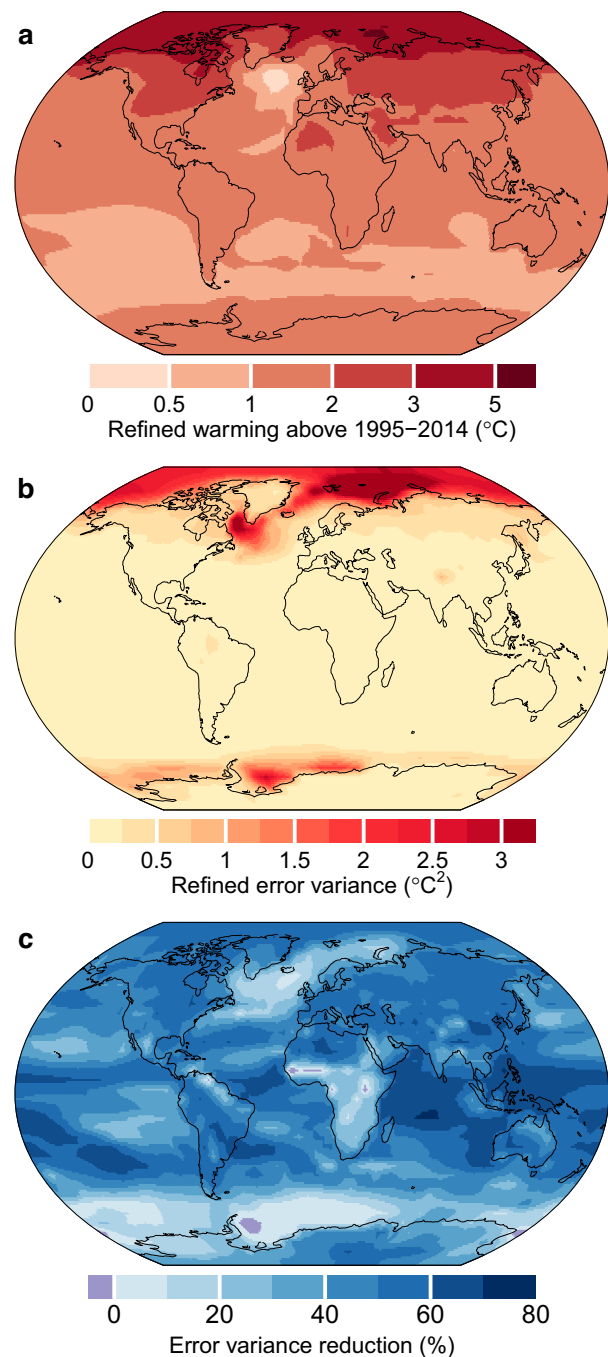


Fig. 4 | Local warming projections refined by constrained global warming. **a, b** Multi-model ensemble mean (**a**) and error variance (**b**) of projected changes in annual mean near-surface air temperature in the mid-term period 2041–2060 relative to the present-day period (1995–2014) under the Shared Socioeconomic Pathway 3-7.0 scenario refined by constrained global warming projections. **c** Percentage reductions in the refined projection error variance relative to the raw value.

Fig. S8), which is not central to the machine learning constraint relationships (Fig. 3a). Thus, differences in SST patterns over the eastern tropical Pacific between models and observations exert minimal influence on our constrained projections.

A distinct feature of the warming field constraint is its effectiveness in raising the lower uncertainty bound of future warming projections. Comparison of observed and simulated warming fields over 1971–2020 helps elucidate the underlying mechanism. Observed grid-

cell warming trends fall within the 5–95% range spanned by model simulations over nearly all locations (Supplementary Fig. S2), and within the more central 16–84% range over the vast majority of grid cells (Supplementary Fig. S9), indicating that the constrained projections are grounded in relationships represented within the model ensemble. Moreover, grid cells with observed trends above and below the simulation median contribute comparably in the learned constraint relationships (Supplementary Fig. S10a–e), preventing the constrained warming from being anchored near the edge of the learned constraint relationship and allowing uncertainty reduction to act on both lower and upper bounds. By contrast, the observed global mean trend lies below the simulation median (Supplementary Fig. S10f), shifting projections constrained by the HEC global mean approach toward lower values and limiting its effectiveness in raising the lower uncertainty bound.

Previous studies have used observed global mean warming to constrain the transient climate response and equilibrium climate sensitivity^{18,20,21,48,67}. Compared with constraints based solely on the global mean, the warming field constraint samples both relatively strong and weak historical warming across regions, reducing reliance on a single aggregated constraint metric and yielding a more balanced reduction of uncertainty in future warming projections. By extension, spatial patterns of historical warming are also likely to contain complementary information relevant to constraining climate sensitivity. In this sense, the machine learning warming field approach points to a promising pathway for future work exploring how the observed spatial warming information may be integrated to refine the understanding of climate sensitivity, including potential implications for its lower uncertainty bound.

Our uncertainty reductions are based on the observed warming field during 1971–2020, a period with relatively more reliable observations, more complete coverage in space and time, and warming dominated by anthropogenic greenhouse gas forcing. Recent studies suggest that the full trajectory of global mean temperature since the preindustrial era offers a stronger constraint on future warming than recent trends alone²³. Additional constraint information may also reside in the long-term memory of temperature variability, which reflects the persistence of climate responses to external forcing and feedbacks over extended timescales^{68–70}. Cloud-related properties have also been identified as effective constraints on future warming⁷¹. Extending warming field observations to earlier periods and including additional process-relevant observables could allow a more complete analysis of the relationship between historical and future temperature variations, thereby further improving constraint strength.

Overall, leveraging machine learning to extract information from the observed global warming field using relations learned from climate models demonstrates a powerful emergent constraint framework for improving future climate projections. Such an approach not only substantially reduces projection uncertainties compared to constraints based on the global mean warming alone, but also reveals key regions where observations most effectively constrain future warming, with clear physical interpretations. Because many climate impacts scale closely with global temperature change, these improved global warming projections can translate into more reliable projections of other climate impact variables for impact assessments. The key regions also highlight where improved temperature monitoring would most benefit future observational constraints.

Methods

The machine learning warming field constraint

We use CMIP6 climate model simulations of near-surface air temperature for historical periods 1850–1900 and 1971–2020, as well as for future near-term (2021–2040), mid-term (2041–2060), and long-term (2081–2100) periods under the SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-

8.5 emissions scenarios⁴⁷. All available simulations from each climate model under each scenario are included, comprising 236 simulations from 45 models for SSP1-2.6, 294 simulations from 46 models for SSP2-4.5, 236 simulations from 39 models for SSP3-7.0, and 277 simulations from 48 models for SSP5-8.5 (Supplementary Table S1). All simulations are bilinearly interpolated to a common 5°×5° grid consistent with the HadCRUT5 observational dataset⁴⁹. For each simulation, we calculate the 1971–2020 historical warming trend at individual grid cells and the area-weighted GSAT warming relative to the 1850–1900 baseline for each future period.

We adopt a multi-output formulation of the Categorical Boosting (CatBoost) algorithm^{72,73} to learn an emergent constraint relationship for jointly estimating GSAT warming for the three future periods under each emissions scenario, using area-weighted 1971–2020 warming trends at individual grid cells ($36 \times 72 = 2592$ grid-cell trends) as input. Training a single model to constrain multiple projection periods maintains model parsimony relative to using separate models for each period, while exploiting the physical coherence of global warming across time horizons under a given emissions pathway. Intuitively, CatBoost identifies combinations of grid cells whose historical warming best predicts future global mean warming by iteratively constructing and refining a sequence of decision trees (i.e., prediction models), with each tree reducing residual errors from the previous one until prediction accuracy converges. To mitigate overfitting, CatBoost constrains how decision trees combine grid-cell trends as predictors through a structured decision-tree framework⁷², promoting robust generalization across simulations from climate models with diverse structures. Its built-in regularization further facilitates the extraction of reliable constraint relationships from high-dimensional warming fields with strong spatial dependence.

Several hyperparameters that influence CatBoost generalization and accuracy must be specified, including the depth of decision trees, which controls the overall model complexity; the learning rate, which determines the contribution of each tree to the prediction; and the regularization coefficient, which controls the strength of regularization used to prevent overfitting. Together, they regulate how fast the model learns and how complex it can become. We tune these hyperparameters using a trial-and-error search (Supplementary Table S4). We find that both the constrained global warming projections and the regions identified as most critical to the constraint relationships are largely insensitive to the specific hyperparameter choices (Supplementary Fig. S11). CatBoost training also involves stochastic components, such as random permutations of the training data during model construction, that rely on the specification of a random seed. Although the constrained projections show only weak sensitivity to the choice of random seed, we nevertheless minimize stochastic variability by using the ensemble mean of 100 constrained estimates obtained from an ensemble of 100 trained CatBoost models, each with a different random seed.

We train CatBoost using all available simulations for each scenario, allowing it to learn constraint relationships from a large collection of forced temperature responses in the presence of internal variability across a wide range of climate sensitivities. While pooling all simulations could, in principle, overweight models with large ensembles, subsampling these models to the multi-model median ensemble size does not materially alter the constrained projections and identified critical regions (Supplementary Fig. S12). The learned constraint relationships can be confidently applied to observations to generate observation-constrained warming projections, provided that the observed warming field reflects characteristics of the Earth system captured within the range of available simulations. This condition is supported by the fact that the 1971–2020 warming trends from the HadCRUT5 [49] observational dataset fall within the 5–95% range of simulated values over nearly all grid cells (Supplementary Fig. S1a) and within the more central 16–84% range over the vast majority

(Supplementary Fig. S9), implying that the constrained projections are grounded in relationships represented within the model ensemble.

We apply the learned constraint relationships to the area-weighted 1971–2020 warming trend maps from four observational datasets, HadCRUT5 [49], NOAA GlobalTemp v6 [51], NASA GISTEMPv4 [52], and CMSTv2 [53] to generate observation-constrained global warming projections. The grid-cell warming trends across these datasets agree closely, with global mean absolute differences of only 0.03 to 0.05 °C per decade and spatial pattern correlations ranging from 0.89 to 0.96. Thus, the constrained projections differ little among these datasets (Supplementary Fig. S1), and we therefore present results constrained by HadCRUT5 in the main text. Because CatBoost does not natively provide uncertainty estimates for its predictions, we characterize uncertainty in the constrained warming projections using an out-of-sample cross-validation framework (described below), a pragmatic approach commonly used in regression models that do not directly estimate predictive uncertainty^{74,75}. Future work could explore probabilistic or Bayesian machine learning frameworks that directly estimate constrained uncertainty.

Cross-validation analysis

We conduct leave-one-model-out cross-validation to verify that the machine learning warming field constraint provides more accurate projections without being overconfident. For each emissions scenario, simulations from each climate model are withheld in sequence as a validation sample (i.e., pseudo-observations of past and future warming). The CatBoost model, using fixed hyperparameters (Supplementary Table S4), is trained on the simulations from the other climate models, and the individual historical warming fields from the withheld simulations are used to generate a corresponding set of constrained warming estimates of that model's future warming. This process is repeated iteratively for each model in turn, resulting in a set of constrained warming estimates.

We compare the machine learning warming field constraint with the HEC global mean constraint, which has previously been shown to effectively reduce uncertainty in global warming projections^{13,18,19,22,71}. The HEC framework is typically used to derive emergent constraint relationships, and thus constrained warming projections, using model ensemble means rather than individual simulations^{13,18,19,22,25,28,29}. To ensure a fair comparison with HEC, we accordingly average the constrained estimates and corresponding pseudo-observations across ensemble members for each model, yielding one constrained warming estimate and one pseudo-observation per model.

We compute the root mean squared error (RMSE) between the constrained estimates and the corresponding pseudo-observations for each climate model. To derive the constrained confidence intervals for each model's warming estimate, we use empirical quantiles of out-of-sample model residuals, defined as the differences between pseudo-observations and constrained estimates from all other models, ensuring that a model's own residuals do not inform its intervals. For example, the 5–95% uncertainty bound for a given model is obtained by adding the 5th and 95th quantiles of residuals from the other models to the model's ensemble mean constrained projection. The coverage rate, defined as the proportion of pseudo-observations falling within the estimated intervals, is compared to the specified nominal level, such as 90% for the 5–95% uncertainty range. Consistency indicates that the constrained confidence intervals are not overly wide or too narrow. For the observation-constrained warming projections (e.g., Fig. 1a), we derive confidence intervals using residuals from all individual simulations.

Understanding the warming field constraints

To understand why the spatially resolved warming field provides additional constraint information beyond the global mean trend, we analyze local warming sensitivities in climate models, defined as the

ratio of local warming to global mean warming. These sensitivities are computed for both the 1971–2020 historical period and future periods. We calculate their ensemble means and standard deviations to characterize their spatial patterns and quantify associated uncertainties. We then assess whether patterns of local warming sensitivities provide meaningful information about future sensitivities by evaluating their temporal stability.

We next show that uncertainty in future local warming sensitivities affects global mean warming projections. To evaluate this, we perform an empirical orthogonal function (EOF) analysis on anomalies of grid-cell warming projections for each projection period and scenario, with anomalies defined relative to the respective ensemble mean projection. In all cases, the first EOF captures the dominant inter-model uncertainty mode in future warming (Supplementary Fig. S3b), and alone explains more than 95% of the projection error variance in global mean warming (Supplementary Fig. S3c). Moreover, its spatial structure closely resembles the pattern of uncertainty in projected local warming sensitivities (Supplementary Fig. S3a, S3b). This correspondence indicates that uncertainty in future local warming sensitivities is a critical contributor to uncertainty in global mean warming projections.

Given the overall stability of local warming sensitivity patterns and their strong influence on global mean warming projection uncertainty, historical sensitivities are expected to be informative of future warming responses. We test this expectation by examining inter-model correlations between historical local warming sensitivities and future global mean warming projections, and by evaluating whether regions exhibiting strong sensitivity-warming correlation coincide with regions where historical local warming trends are strongly correlated with future global warming projections.

Interpreting the machine learning results

To assess the relative importance of historical warming trends (and sensitivities) at individual grid cells in constraining future global warming, we use the CatBoost Prediction Value Change (PVC) metric⁷², which quantifies how much the constrained warming changes (in °C) when a grid cell is included or excluded. We normalize PVC values across grid cells so that their area-weighted sum equals 100%. The normalized values thus indicate the relative importance of individual grid cells in the learned constraint relationships and the constrained warming projections. To mitigate stochastic variability arising from random components in model training, we compute ensemble mean normalized PVC values across 100 trained CatBoost models, each initialized with a different random seed. For constrained projections under each emissions scenario, we then identify the 100 most influential grid cells (Supplementary Fig. S4), which together account for approximately 57, 81, 81, and 82% of the total influence under SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5 scenarios, respectively, while the remaining 2492 grid cells each contribute only marginally on an individual basis. To highlight locations that are consistently influential across scenarios, we additionally identify the 100 most influential grid cells based on PVC values averaged across emissions scenarios (Fig. 3a).

An independent physically informed interpretation

For a set of climate models in which temperature sensitivities to effective radiative forcing remain approximately constant with warming level, the forced warming response to a given effective radiative forcing F (in W m^{-2}) can be approximated, to first order, by a random variable $F \times TS$, where TS denotes the modeled temperature sensitivity per unit forcing (that is, a Green's function representation neglecting slow, state-dependent responses⁷⁶). Within this framework, the inter-model standard deviation of simulated warming responses is expected to scale linearly with the magnitude of the applied forcing. Assuming that the influence of internal climate variability on long-term

warming projections is small, the ratio of the inter-model standard deviation of future warming projections driven by forcing F_F to that of historical simulations driven by F_H is therefore approximately given by F_F/F_H . We use this simple scaling relationship to provide an independent, physically grounded interpretation of how CatBoost identifies the most influential grid cells.

We estimate the effective radiative forcing $F(t)$ driving historical and future GSAT changes as follows^{76,77}: $F(t) = \Delta N(t) + \lambda \Delta \text{GSAT}(t)$, where $\Delta N(t)$ is the global mean top-of-atmosphere energy imbalance diagnosed from the required model radiation outputs, $\Delta \text{GSAT}(t)$ denotes GSAT changes relative to 1850–1900, and λ is the model-specific climate feedback parameter taken from ref. 67. Effective radiative forcing is computed for each simulation, averaged across simulations within each model, and then across models. The resulting multi-model ensemble mean estimates are used to calculate the ratio of effective radiative forcing driving historical and future warming. Due to the limited availability of the required radiation outputs, fewer climate models are included in the effective radiative forcing analysis than in the warming projections (Supplementary Table S5).

To evaluate consistency with the expected linear forcing-response scaling at the grid-cell level, we focus on the dominant modes of uncertainty in historical and future warming. We apply EOF analysis to the 1971–2020 warming trend simulations and future warming projections and extract the dominant first EOF mode. Grid-cell warming amounts are then filtered to isolate the contributions from this dominant mode by regressing the associated principal component onto grid-cell warming simulations. The inter-model standard deviation of these EOF-filtered warming amounts is subsequently calculated at each grid cell for both historical and future periods, and then used to assess whether the grid cells identified as most influential by the CatBoost model are consistent with expectations from the linear forcing-response scaling relationship.

Refining local climate projections using constrained global warming

We combine the constrained global warming projections with the level-of-global-warming approach^{13,59} to constrain projections of local climate change. For a given period and emissions scenario, global mean warming projections from an ensemble of climate models are adjusted to conform with the constrained distribution using quantile mapping, ensuring that the raw and adjusted projections correspond to the same quantiles in their respective distributions. Model time windows during which models reach their adjusted global warming projections are identified based on their raw global mean warming projections. Grid-cell climate change projections within these time windows represent the projections constrained by and thus are consistent with the constrained global warming projections.

Data availability

The observational near-surface air temperature datasets HadCRUT5, NOAA GlobalTemp v6, NASA GISTEMPv4, and CMSTv2 are available at <https://www.metoffice.gov.uk/hadobs/hadcrut5/>, <https://www.ncei.noaa.gov/products/land-based-station/noaa-global-temp>, <https://data.giss.nasa.gov/gistemp/>, and <https://www.gwpu.net>, respectively. The Reconstructed Sea Surface Temperature version 5 dataset is available at <https://www.ncei.noaa.gov/products/extended-reconstructed-sst>. The CMIP6 climate model simulations are available through the ESGF data portal at <https://esgf-node.llnl.gov/search/cmip6/>.

Code availability

The machine learning models are implemented using the Python packages catboost version 1.2.8 (<https://pypi.org/project/catboost>) and xgboost version 2.1.3 (https://xgboost.readthedocs.io/en/stable/python/python_intro.html). Other analyses in this study do not rely on

specific codes and can be reproduced following the descriptions in the Methods section.

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Author contributions

C.L. conceived of and designed the study. J.W. and X.C. conducted the machine learning analysis; Z.W. prepared the data and performed the hierarchical emergent constraint analysis. C.L., F.W.Z., and X.Z. developed the physical interpretation of the machine learning results and wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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